



Figure 1: The TensorFlow logo

Deep learning can be used for various real world applications including speech recognition, malware detection and classification, natural language processing, bioinformatics, computer vision and many others.

TensorFlow: A Python based open source software library for deep learning

TensorFlow ([tensorflow.org](https://www.tensorflow.org)) is a powerful open source software library for the implementation of deep learning. It is based on Python and C++ at the back-end, with the incorporation of algorithms for data flow as well as graphs based numerical computations to achieve multi-layered computations with higher accuracy and a lower error rate.

TensorFlow has been developed by Google under a research project for deep learning titled Google Brain. It is a second generation system developed by Google after DistBelief. TensorFlow was released as open source in November 2015 by Google and this move has motivated research scholars, academicians and scientists to work on this powerful library.

TensorFlow can be installed with binary packages or authorised GitHub sources. Any one of the following methods can be used to install it:

- Python Pip
- Virtualenv
- Anaconda
- Docker Toolbox
- Source based installation

Running and testing TensorFlow on the command line

At the terminal, the following instructions can be executed for testing:

```
$ python
>>> import tensorflow as myTensorFlow
>>> MyMessage = myTensorFlow.constant('Hello To All')
>>> mysession = myTensorFlow.Session()
>>> print(sess.run(MyMessage))
Hello To All
>>> x = myTensorFlow.constant(9)
```

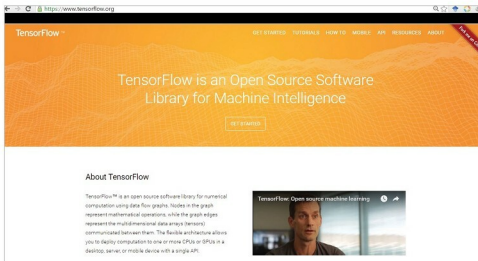


Figure 2: The official TensorFlow page

```
>>> y = myTensorFlow.constant(2)
>>> print(mysession.run(x + y))
11
>>>
```

Using TensorFlow for network packet analysis

For classification and prediction of malware from network packets, the dataset should be fetched from any authentic source like honeypots, honeynets, data repositories, open data portals, etc. Many anti-virus companies and research organisations release their datasets for R&D purposes, enabling research algorithms to be implemented by research scholars and practitioners.

If the datasets of malware or network traffic are required, the following URLs can be used:

- <http://www.netresec.com/?page=PCapFiles>
- <https://wiki.wireshark.org/SampleCaptures>
- <https://snap.stanford.edu/data/#email>
- <http://www.secrepo.com/>

As the network traffic is captured in PCAP format, there is a need to transform the PCAP format to CSV (comma separated values) using Snort IDS, by which all the alert files from PCAP can be generated and then changed to CSV format.

To read a pcap file in Snort IDS, use the following code:

```
$ snort -r mypcap.pcap
$ snort --pcap-single= mypcap.pcap
```

The alert file format generated by Snort IDS is as follows:

```
["[" [1:2016936:2] Suspicious inbound to Database port 1433
["["
[Classification: Potentially Bad Traffic] [Priority: 2]
07/07-10:10.817217 <IP>:<PORT> -> <IP>:<PORT>
TCP TTL:112 TOS:0x0 ID:256 IpLen:24 DgmLen:20
```